

AUTOCATALYTIC REACTIONS IN THE ISOTHERMAL, CONTINUOUS STIRRED TANK REACTOR

OSCILLATIONS AND INSTABILITIES IN THE SYSTEM $A + 2B \rightarrow 3B$; $B \rightarrow C$

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Abstract—The prototype, cubic autocatalytic reaction ($A + 2B \rightarrow 3B$) forms the basis for the simplest homogeneous system to display "exotic" behaviour. Even under well-stirred, isothermal, open conditions (CSTR) we may find multistability, hysteresis, extinction, ignition and anomalous relaxation times. Allowing for the finite lifetime of the catalyst ($B \rightarrow$ inert products) adds another dimension. The dependence of the stationary-states on residence-time now yields isolas or mushrooms. Sustained oscillations (stable limit cycles) are also possible. The onset of oscillation corresponds to a change in the character of the stationary-state (from stable focus to unstable focus) and the conditions for this change can be evaluated analytically. The period of the oscillations and their amplitudes increase as the residence time is lengthened. A total of nine different phase-portraits in the a - b plane has been found.

This isothermal system is less complex than the exothermic, first-order reaction in a CSTR, but there are strong analogies between the two.

1. INTRODUCTION

This paper is intended to be read in conjunction with our recent investigation[1] of isothermal autocatalytic reactions in a continuous-flow well-stirred tank reactor (CSTR). The latter considered the two simple, prototype schemes:

quadratic	$A + B \rightarrow 2B$	rate = kab
cubic	$A + 2B \rightarrow 3B$	rate = kab^2 .

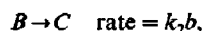
Even when the catalyst B is indefinitely stable, these show ranges of residence-time over which there are multiple stationary-states and ranges of unique solutions.

For the *quadratic rate-law*, the effect of including some of the autocatalyst B in the inflow is to remove multiplicity.

For *cubic autocatalysis*, multiplicity survives provided the concentration of B in the inlet b_0 is not too high. However, at all long residence-times all solutions become unique. The range over which there are three stationary-states decreases as b_0 increases. Multiplicity vanishes for $b_0 \geq a_0/8$. Unique solutions are always stable; when there are three, the uppermost and the lowest are stable and the middle one is always unstable. Any small perturbation of the system away from a stable stationary-state, decays monotonically and, in general, exponentially. The relaxation time for this return may become very long compared with the residence time, and the decay then becomes

non-exponential. There has been earlier, related work on these features by Lin[2] and by Heinrichs and Schneider[3].

The two schemes above are one-variable systems, because the concentrations of A and B are not independent, but are uniquely related to the inlet composition by the reaction stoichiometry ($A \rightarrow B$). In circumstances where the autocatalyst B is not indefinitely stable, but undergoes a first order decay (or poisoning)



the behaviour is even richer. When there is no catalyst present in the inflow ($b_0 = 0$) the range of multiple stationary-states appears as a closed curve or isola (see, e.g. Fig. 1). This lies above the lowest solution,

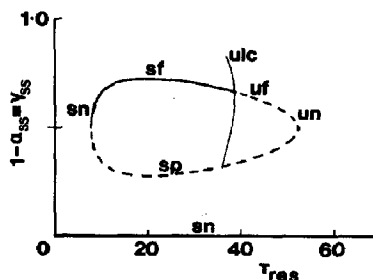


Fig. 1. Dependence of stationary-state extent of conversion $\gamma_{ss} = 1 - \alpha_{ss}$ on residence-time τ_{res} for cubic autocatalysis with decay ($\tau_2 = 20$) and no catalyst in inflow ($\beta_0 = 0$): isola formation: sn, stable node; sf, stable focus; un, unstable node; uf, unstable focus; sp, saddle point; ulc, unstable limit cycle; slc, stable limit cycle.

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which exists for all residence-times and corresponds to no conversion of A to B . If some autocatalyst is added to the feed, the lowest branch rises from the axis, attaining a maximum at $t_{res} = 1/k_2$. If this maximum is relatively low, we may still find an isola above it. As the inlet concentration of B is increased, the lower and middle solution (which forms the bottom of the isola) coalesce. The closed curve opens up to give a mushroom (Figs. 2 and 3).

We have also studied elsewhere the effects on the patterns of stationary-states: (i) of reversibility of the autocatalytic reaction[4]; and (ii) of a simultaneous, uncatalysed reaction[5] converting A to B .

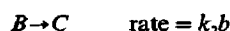
In this paper, we shall present our investigations of the stability and character of the different stationary-state. When no catalyst is included in the inflow, the lowest solution (i.e. that corresponding to the lowest extent of conversion) is always a stable node. Small

perturbations decay monotonically. When the inlet concentration of B is non-zero, this stationary-state may become unstable. The middle solution is always unstable: it is a saddle point. The character of the highest stationary-state (greatest extent of conversion) may be nodal or focal (perturbations decay via damped oscillations) and this solution may be stable or unstable. When it is unstable and unique, e.g. on the upper branch of the mushroom in Fig. 3, sustained oscillations arise, with a stable limit cycle in the a - b phase plane. Unstable limit cycles surrounding stable foci are also possible. Finally limit cycles that surround all three stationary-states can be found.

We shall also discuss the strong analogies between our model and the first-order, irreversible, exothermic reaction proceeding under non-isothermal conditions in a CSTR. The isothermal, autocatalytic reaction is simpler, yielding to algebraic analysis. Systems with a completely stable catalyst ($k_2 = 0$) resemble adiabatic operation of the non-isothermal case: isothermal systems with a catalyst that decays resemble non-adiabatic operation. The non-isothermal system, however, can display a wider range of patterns for the dependence of stationary-states on residence-time or flow rate.

2. MASS BALANCE EQUATIONS AND STATIONARY-STATES

We shall consider throughout the two reaction scheme



where C is some inert product. The mass-balance equations, which relate the rates of the chemical steps to the inflow and outflow rates, are

$$\frac{da}{dt} = -k_1ab^2 + k_f(a_0 - a) \quad (1)$$

$$\frac{db}{dt} = k_1ab^2 - k_2b - k_f(b - b_0).$$

Here k_f is the inverse of the mean residence time t_{res} and a_0 and b_0 are the concentrations of A and B in the inlet stream.

It is convenient for the present purposes to use separate, dimensionless measures of the two species concentrations. In our previous paper, we were mainly concerned with the stationary-states of (1). The condition for a stationary-state provides an additional constraint on the system, which can then be described fully by a single species concentration. Let

$$\alpha = a/a_0 \quad \text{and} \quad \beta = b/a_0. \quad (2)$$

The extent of conversion γ used previously is given by $\gamma = 1 - \alpha$.

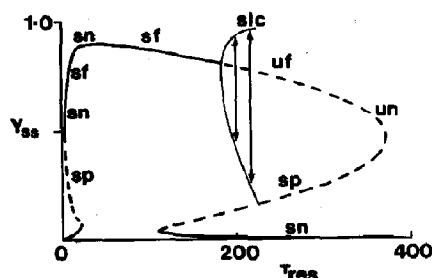
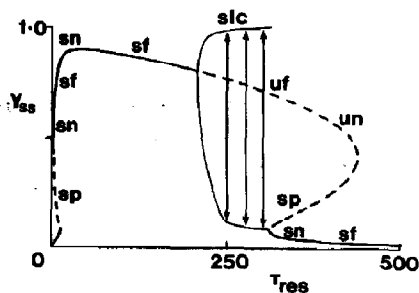
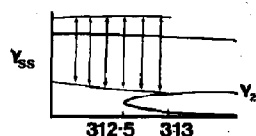


Fig. 2. Dependence of stationary-state extent of conversion $\gamma_{ss} = 1 - \alpha_{ss}$ on residence-time for cubic autocatalysis with decay ($\tau_2 = 40$) and some catalyst in inflow ($\beta_0 = 1/30$): mushroom formation.



(a)



(b)

Fig. 3. (a) Dependence of stationary-state extent of conversion $\gamma_{ss} = 1 - \alpha_{ss}$ on residence-time for cubic autocatalysis with decay ($\tau_2 = 40$) and some catalyst in inflow ($\beta_0 = 1/15$): mushroom formation. (b) Enlargement in vicinity of $\tau_{res} = 313$ showing reappearance of α_1 and α_2 with respect to amplitude of sustained oscillations about α_3 .

A convenient characteristic time-scale t_{ch} for time-dependent features also emerges naturally. This is based on the rate of our principal, autocatalytic reaction step,

$$t_{ch} = 1/k_1 a_0^2. \quad (3)$$

This differs by a factor of 4/27 from the characteristic time most appropriate for discussing stationary-states (see [1] for a full discussion). Other choices for t_{ch} based on k_2 or k_f , are also possible, but (3) leads to much neater forms and to greater clarity of the effects of residence-time and the catalyst decay rate. In these terms, eqn (1) becomes

$$\begin{aligned} \frac{d\alpha}{d\tau} &= -\alpha\beta^2 + \frac{(1-\alpha)}{\tau_{res}} \\ \frac{d\beta}{d\tau} &= \alpha\beta^2 - \frac{\beta}{\tau_2} - \frac{(\beta-\beta_0)}{\tau_{res}} \end{aligned} \quad (4)$$

where $\beta_0 = b_0/a_0$, dimensionless catalyst inflow; $\tau = t/t_{ch} = k_1 a_0^2 t$, dimensionless time; $\tau_{res}/t_{ch} = k_1 a_0^2/k_f$, dimensionless residence time; $\tau_2 = t_2/t_{ch} = k_1 a_0^2/k_2$, dimensionless decay time.

Stationary-states of (4) arise when $d\alpha/d\tau = d\beta/d\tau = 0$. The concentrations of α and β are then strictly related

$$\beta_{ss} = (1 + \beta_0 - \alpha_{ss})/[1 + (\tau_{res}/\tau_2)]. \quad (5)$$

This relationship reduces the dimension of the system to one. In the limit $\tau_2 \rightarrow \infty$ (indefinitely stable catalyst), this reduces to

$$\beta_{ss} - \beta_0 = 1 - \alpha_{ss} = \gamma_{ss},$$

i.e. the dimensionless increase in the concentration of autocatalyst B above its inlet value is exactly equal to the dimensionless extent of conversion of A . For a catalyst of finite lifetime, this equality does not hold, because some B is subsequently converted to the product C .

The solutions α_{ss} , lying in the range $0 \leq \alpha \leq 1$, are given by the roots of the cubic equation

$$\begin{aligned} \alpha_{ss}^3 - 2(1 + \beta_0)\alpha_{ss}^2 + \left[(1 + \beta_0)^2 + \frac{1}{\tau_{res}} \right. \\ \left. + \left(1 + \frac{\tau_{res}}{\tau_2} \right)^2 \right] \alpha_{ss} - \frac{1}{\tau_{res}} \left(1 + \frac{\tau_{res}}{\tau_2} \right)^2 = 0. \end{aligned} \quad (6)$$

These are readily evaluated numerically for any given values of β_0 , τ_2 and τ_{res} . There will be either one or three real solutions. Figures 1–3 display three such examples of the dependence of the stationary-state extent of conversion $\gamma_{ss} = 1 - \alpha_{ss}$ on the residence time, for different choices of the inlet concentration of B and its decay rate. High values of α_{ss} , close to unity, correspond to small extents of reaction: low values indicate large extents.

Special case: inflow pure A

If there is no autocatalyst B in the inflow (i.e. if $\beta_0 = 0$), one solution of (6) corresponds to no conversion; $\alpha_{ss} = \alpha_1 = 1$, $\beta_{ss} = 0$. The other two stationary-states α_2 and α_3 have $\alpha_{ss} \neq 1$. They are given by the roots of the equation

$$\alpha_{ss}(1 - \alpha_{ss}) = \frac{1}{\tau_{res}} \left(1 + \frac{\tau_{res}}{\tau_2} \right)^2. \quad (7)$$

This is quadratic both in α_{ss} and in τ_{res} . Again the stationary-state concentrations of B are related to α_{ss} by (5).

The condition for multiple solutions in these circumstances ($\beta_0 = 0$) is clearly then

$$\frac{4}{\tau_{res}} \left(1 + \frac{\tau_{res}}{\tau_2} \right)^2 \leq 1.$$

This inequality can only be satisfied if

$$\tau_2 \geq 16.$$

When this is so, α_2 and α_3 lie on a closed curve or isola extending over some range of the residence time τ_{res} , as shown in Fig. 1.

3. LOCAL STABILITY OF STATIONARY-STATES

The character and local stability of each of the three possible stationary-states at any given residence time, $\alpha_1 \geq \alpha_2 \geq \alpha_3$, are determined by the eigenvalues λ_1 and λ_2 of the Jacobian matrix evaluated at the appropriate solution. These are given by the roots of the quadratic equation

$$\begin{vmatrix} \frac{\partial}{\partial \alpha} \left(\frac{d\alpha}{d\tau} \right) - \lambda & \frac{\partial}{\partial \beta} \left(\frac{d\alpha}{d\tau} \right) \\ \frac{\partial}{\partial \alpha} \left(\frac{d\beta}{d\tau} \right) & \frac{\partial}{\partial \beta} \left(\frac{d\beta}{d\tau} \right) - \lambda \end{vmatrix}_{ss} = 0.$$

3.1 No catalyst in inflow ($b_0 = 0$)

For the special case in which there is no catalyst in the inflow, the solution α_1 , which corresponds to the lowest stationary-state extent of conversion γ_1 in Fig. 1, is always stable. Both roots λ are negative. They are also always real, with no imaginary parts. Thus α_1 is approached monotonically. It is a stable node (*sn*). The character and stability of this solution is unaffected by the existence or non-existence of the other stationary-states α_2 and α_3 . When there are multiple stationary-states, the middle one α_2 is always found to be an unstable point. The roots λ_1 and λ_2 for this middle solution have opposite signs. This is a saddle point and is not realizable in practice.

The stability and character of the highest extent of conversion γ_3 (corresponding to the lowest stationary-state concentration of A , α_3) varies considerably. It is not the same at all residence times. The exponents λ_1 and λ_2 may be real or complex and may be positive or

negative. They depend strongly on τ_{res} . Table 1 summarizes the different possibilities for the special case in which there is no catalyst in the inflow. The different natures of $\gamma_3 = 1 - \alpha_3$ are also indicated on the appropriate branches in Fig. 1.

3.2 Systems with catalyst in the inflow ($b_0 > 0$)

When the concentration of the catalyst in the inflow is not zero, it is possible to find a wider variety in the character and stability of the lowest stationary-state a_1 or γ_1 . This solution may be either a node or a focus, and may become unstable. This instability only arises for systems which display mushrooms, and occurs along the branch of solutions lying at long residence times. The branch of stationary-states at the shortest residence-times (i.e. to the left of the mushroom—the so-called “flow-branch”) is always stable.

The middle solution is still always a saddle point. The upper branch of stationary-states may still display stability or instability and nodal or focal character.

These analysis yield only local information about the stationary-states—or singular points in the a - b phase plane. To complete these maps, we require additional information which reveals the location and stability of limit cycles.

We know from the theorems of Hopf, that such closed trajectories come into being in the vicinity of the change from stable focus to unstable focus. The limit cycles will be unstable if they surround the stable branch of solution; they will be stable if they surround unstable foci. Our purpose in the remaining sections is to evaluate the range of residence times over which limit cycles surround the upper stationary-state γ_3 of our system and to assess their

stability. If they are stable, we wish in addition to calculate the amplitude and frequency of the resulting sustained oscillations.

4. OSCILLATIONS: LIMIT CYCLES IN a - b PHASE PLANE

There are two main routes to obtaining the period, amplitude and stability of oscillations exhibited by a system of differential equations. The first is analytical, but of limited scope: the second is computational, and presents a vast array of numerical results that must be digested. In the present problem these constitute complementary approaches, each being easiest to apply under circumstances where the other is least manageable.

4.1 Hopf bifurcation: analytical solutions

An algebraic analysis of the oscillatory solutions of (4) is easy and practical only if the inlet concentration of B is taken to be zero (i.e. if $\beta_0 = 0$). When this is so, we can express α_3 and β_3 explicitly in terms of τ_{res} and τ_2 :

$$\alpha_3 = \frac{1}{2} \{1 - (1 - 4T)^{1/2}\};$$

$$\beta_3 = \frac{1}{2} \{1 + (1 - 4T)^{1/2}\} / (1 + \tau_{res}/\tau_2)$$

where

$$T = \frac{1}{\tau_{res}} \left(1 + \frac{\tau_{res}}{\tau_2}\right)^2.$$

This group T is proportional to $1/\tau_{res}$ when τ_{res}/τ_2 is small and to τ_{res} when τ_{res}/τ_2 is large. It passes through a minimum when $\tau_{res} = \tau_2$; $T_{min} = 4/\tau_2$.

Table 1. Stability and character of stationary-state solutions as functions of residence time τ_{res} and catalyst decay rate τ_2 expressed in terms of concentration of catalyst B ; $\beta_{ss} = [1 + (1 - 4T)^{1/2}]/2[1 + (\tau_{res}/\tau_2)]$. $T = [1 + (\tau_{res}/\tau_2)]^2/\tau_{res}$. No catalyst in inflow ($\beta_0 = 0$): isola

stationary state	range of β_{ss} ($\kappa = \tau_{res}/\tau_2$)	character and stability	
		$\tau_{res} < \tau_2$ ($\kappa < 1$)	$\tau_{res} > \tau_2$ ($\kappa > 1$)
lowest	$\beta_{ss}^2 \tau_{res} = 0$	stable node	stable node
middle	$0 < \beta_{ss} \tau_{res} < 1$	saddle point	saddle point
	$1 < \beta_{ss}^2 \tau_{res} < \{[2(1 + \kappa)]^{1/2} - \kappa^{1/2}\}^2$	stable node	unstable node
highest	$\{[2(1 + \kappa)]^{1/2} - \kappa^{1/2}\}^2 < \beta_{ss}^2 \tau_{res} < \kappa$	stable focus	unstable focus
	$\kappa < \beta_{ss}^2 \tau_{res} < \{[2(1 + \kappa)]^{1/2} + \kappa^{1/2}\}^2$	stable focus	stable focus
	$\{[2(1 + \kappa)]^{1/2} + \kappa^{1/2}\}^2 < \beta_{ss}^2 \tau_{res}$	stable node	stable node

The Jacobian matrix has the form

$$\mathbf{A} = \begin{bmatrix} -\beta^2 - \frac{1}{\tau_{\text{res}}} & -2\alpha\beta \\ \beta^2 & 2\alpha\beta - \left(\frac{1}{\tau_2} + \frac{1}{\tau_{\text{res}}}\right) \end{bmatrix}$$

$$= \frac{1}{\tau_{\text{res}}} \begin{bmatrix} -\frac{[1 + (1 - 4T)^{1/2}]}{2T} & -2\left(1 + \frac{\tau_{\text{res}}}{\tau_2}\right) \\ \frac{[1 - 2T + (1 - 4T)^{1/2}]}{2T} & \left(1 + \frac{\tau_{\text{res}}}{\tau_2}\right) \end{bmatrix}$$

The characteristic equation for λ_1 and λ_2 , is, therefore,

$$\lambda^2 - P\lambda + Q = 0$$

where

$$P = \frac{1}{\tau_2} + \frac{1}{\tau_{\text{res}}} \left\{ 1 - \frac{[1 + (1 - 4T)^{1/2}]}{2T} \right\}$$

and

$$Q = \frac{\tau_2}{2\tau_{\text{res}}(\tau_2 + \tau_{\text{res}})} [1 - 4T + (1 - 4T)^{1/2}].$$

The conditions for Hopf bifurcation, or equivalently for a stable focus to become unstable, is that λ_1 and λ_2 become purely imaginary; i.e., that $P = 0$. This is satisfied when $T = (1/\tau_2)^{1/2}$ or equivalently

$$\tau_{\text{res}} = \tau_{\text{res}}^* = \frac{1}{2} \tau_2 \{ \tau_2^{1/2} - 2 + [\tau_2^{1/2}(\tau_2^{1/2} - 4)]^{1/2} \}.$$

Thus Hopf bifurcation and sustained oscillatory behaviour can only occur in systems for which τ_2 is greater than 16. For $\tau_2 = 16$, $\tau_{\text{res}}^* = 16$; for $\tau_2 = 25$, $\tau_{\text{res}}^* = 65.4$. The residence time at which bifurcation occurs increases rapidly with τ_2 . This is reflected by the elongation of the τ_{res} -axis between Fig. 1 ($\tau_2 = 20$) and 2 ($\tau_2 = 40$).

Further manipulation of the governing equation and various partial differential coefficients evaluated at the stationary-state corresponding to τ_{res}^* , allows the evaluation of the actual time-dependent solutions $\alpha(\tau)$ and $\beta(\tau)$, in terms of a series expansion of $(\tau_{\text{res}} - \tau_{\text{res}}^*)^{1/2}$. These operations are rather involved, but are clearly outlined in general terms in Ref. [6]. We present only the results here, for a particular example. Figure 1 corresponds to a system with $\tau_2 = 20$ and no B in the inflow. From above we calculate $\tau_{\text{res}} = 39.25$. All other conditions required for there to be a Hopf bifurcation at this point are satisfied. Following [6] we find that periodic solutions exist over a range of residence times to the left of τ_{res}^* ($\tau_{\text{res}} < \tau_{\text{res}}^*$). The Floquet exponent for these limit cycles is positive, and hence they are unstable.

For a larger value of the catalyst decay time, $\tau_2 = 40$, we find stable limit cycles branching to the right of $\tau_{\text{res}}^* = 163.2$ (i.e. at higher residence times). The period of these sustained oscillations for a given $\tau_{\text{res}} > \tau_{\text{res}}^*$ is given by

$$p = \frac{2\pi}{\omega_0} [1 + 7.50 (\tau_{\text{res}} - \tau_{\text{res}}^*) + O(\tau_{\text{res}} - \tau_{\text{res}}^*)^2]$$

where ω_0 is the fundamental frequency evaluated at τ_{res}^* : $\omega_0 = Q^{1/2} = 24.2 \times 10^{-3}$. The solutions themselves, apart from an arbitrary phase angle, are

$$\alpha - \alpha^* = \frac{(\tau_{\text{res}} - \tau_{\text{res}}^*)^{1/2}}{10.72} e^{2\pi i t/p} - \frac{(\tau_{\text{res}} - \tau_{\text{res}}^*)}{114.8} [0.77 e^{-4\pi i t/p} - 1.36 e^{4\pi i t/p} + 2.1]$$

$$\beta - \beta^* = -\frac{(\tau_{\text{res}} - \tau_{\text{res}}^*)^{1/2}}{21.44} e^{2\pi i t/p} + \frac{(\tau_{\text{res}} - \tau_{\text{res}}^*)}{114.8} [0.28 e^{-4\pi i t/p} - 0.12 e^{4\pi i t/p} + 0.41]$$

where α^* and β^* are the unstable stationary-states corresponding to τ_{res}^* .

This then shows that, for the model described by eqn (2), even with no autocatalyst in the inflow a supercritical Hopf bifurcation may occur. This leads to *stable limit cycles surrounding unstable foci* and hence to *sustained oscillations over some range of* $\tau_{\text{res}} > \tau_{\text{res}}^*$.

4.2 Numerical computation

Despite its apparent attractions, the analytical approach above is an involved process at best. Both for non-zero values of b_0 and for residence times not very close to τ_{res}^* , it becomes prohibitively difficult. A far more direct route is to choose some arbitrary initial point and to compute the trajectory over a suitable period of time. We have computed such concentration-time histories using a routine based on Gear's method, for various combinations of residence time, catalyst decay rate τ_2 and inlet concentration β_0 . In general this would be an onerous task, but we are guided for the present reaction scheme by the results of our previous paper. Because we already know the pattern of stationary-states as a function of τ_{res} , and their difficult stabilities, we know which points are likely to be of most interest. We present typical results in the next section.

5. COMPUTED RESULTS

(a) Concentration histories

Figures 4–6 show representative traces for three different patterns of response. These have been computed specifically for $\tau_2 = 40$ and $\beta_0 = 1/15$ —see Fig. 3—but are illustrative of quite general behaviour for arbitrary choices of these parameters. Also shown are the corresponding trajectories in the α – β phase plane.

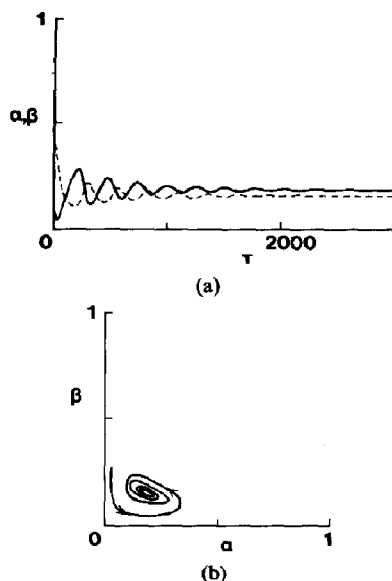


Fig. 4. (a) Decay of small perturbation corresponding to damped oscillations about a stable focus: $\tau_{res} = 190$, $\tau_2 = 40$, $\beta_0 = 1/15$. (b) Phase-plane representation of (a), showing trajectory spiralling onto unique stable, singularity.

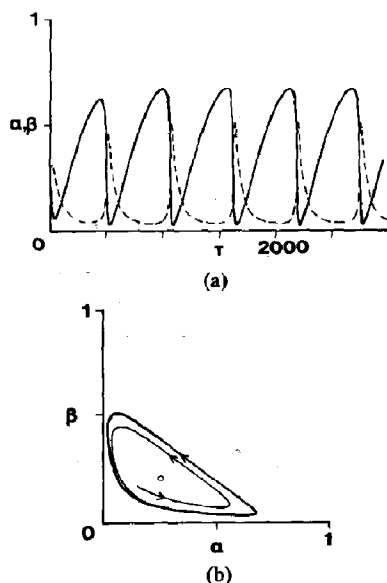


Fig. 5. (a) Sustained, undamped oscillations in concentrations of a (—, α) and B (---, β): $\tau_{res} = 225$, $\tau_2 = 40$, $\beta_0 = 1/15$. (b) Phase-plane representation of (a), showing trajectory unwinding onto stable-limit cycle surrounding unique, unstable singularity.

In Fig. 4, α_3 is a stable focus. The system is initially removed from the singularity, but moves towards it via a series of damped oscillations. The phase-plane shows a trajectory spiralling in to the point α_3, β_3 . Figure 5 shows sustained, undamped oscillations in a and b . These correspond to a stable limit cycle surrounding an unstable focus. After an initial period of

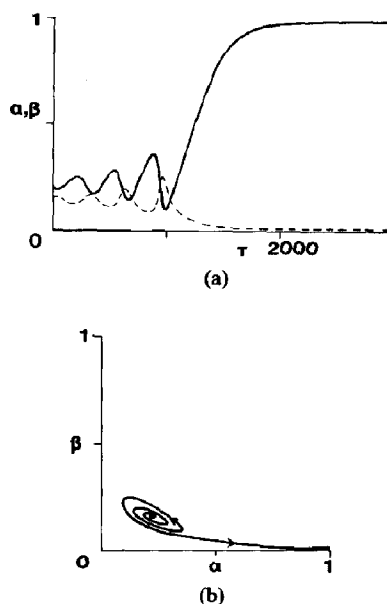


Fig. 6. (a) Divergent oscillations in concentrations of A (—, α) and B (---, β): $\tau_{res} = 315$, $\tau_2 = 40$, $\beta_0 = 1/15$. (b) Phase-plane representation of (a), showing trajectory unwinding from unstable singularity α_3 and crossing α_2 to stable node α_1 .

“settling-down”, regular pulses with constant frequency and amplitude are obtained. In the α - β phase plane the system unwinds onto a closed curve which surrounds the point α_3, β_3 .

For Figs. 4 and 5, α_3 was a unique stationary-state. A third pattern is possible if the other two solutions α_1 and α_2 are also realizable. The oscillations shown in Fig. 6 are divergent. Ultimately the system crosses the separatrices of the saddle point α_2 and settles at the stable node of low reactivity α_1 . In the a - b phase plane, this bifurcation is represented by the merging of a stable limit cycle and a saddle point. This response is found at long residence-times on the α_3 branch in Figs. 2 and 3. These upper solutions are then no longer realizable in practice. No matter how close one starts to α_3 , the system eventually moves to the lower stationary-state α_1 .

It is, however, possible for there to be a stable limit cycle surrounding α_3 , and hence for the system to exhibit sustained oscillations, even if it is not a unique solution and α_2 and α_1 exist. The algebraic analysis for the system with $\beta_0 = 0$ presented in Section 4, shows that the limit cycle is stable over some range of τ_{res} even though the upper stationary-state forms part of an isola. Provided the oscillations are sufficiently small in amplitude that they do not touch the separatrix of the saddle, we may find an oscillatory state that remains in the vicinity of α_3 indefinitely.

(b) Oscillatory frequency and amplitude: influence of residence-time

We may establish how the amplitude and fre-

quency of the oscillations depend on the residence-time by repeating our calculations at various τ_{res} . In Fig. 7 we show that the pulses rapidly become larger but less frequent as the residence-time is increased. Because our oscillations are not sinusoidal we use a slightly more general definition of the amplitude as the difference between the maximum and minimum concentration of A observed during an individual cycle. Thus for $\tau_2 = 40$, $\beta_0 = 1/15$ and $\tau_{res} = 225$, α varies between 0.025 and 0.678 and hence the amplitude of the oscillations is 0.653.

We have also shown the maximum and minimum values of $\gamma = 1 - \alpha$ in Figs. 2 and 3. These indicate the size of the oscillations compared with the location of the different possible stationary-states at any given residence time. The figures reveal clearly how oscillations die out: at short residence times they become vanishingly small, and at long residence times the lower locus touches the branch of saddle points γ_2 . This merging of limit cycle and saddle point is shown more clearly in Fig. 3(b), which expands the γ and τ_{res} scales. The middle stationary-state and the lowest extent of conversion achieved during oscillation do not coalesce at the same residence time at which γ_2 becomes real. They do so at a slightly longer τ_{res} .

For a system with the residence time held constant, the oscillations correspond to motion up and down a line in Figs. 1–3. This line represents the projection onto the α - τ_{res} or γ - τ_{res} plane of the limit cycle which lies in the α - β (or γ - β) plane. The oscillations in time correspond to trajectories around a closed curve perpendicular to the plane of the paper.

(c) Experimental interpretations: hysteresis

The responses which would be observed during an experiment with $\tau_2 = 40$ and $\beta_0 = 1/15$ in which the residence-time is varied from an initial value $\tau_{res} < 200$, would then be as follows (see Fig. 3). Suppose $\tau_{res} = 150$. Then any small initial perturbation will decay back to the unique stable focus in a damped oscillatory manner, similar to that represented in Fig. 4. As τ_{res} is increased, the degree of damping becomes less and any disturbance takes longer to die away. At $\tau_{res} = ca. 200$, the oscillatory response becomes undamped and sustained oscillations will develop. This is called soft excitation. The

system can no longer sit at α_3 , which has become an unstable focus. For such values of the residence-time the α - τ and β - τ traces are approximately sinusoidal. The pulses are small and they are separated by relatively short periods (see Fig. 7). If circumstances are now changed in the reverse direction, i.e. if the residence time is now reduced, the oscillations will die out at precisely the same value of τ_{res} at which they first appeared. There is no hysteresis at the limit.

If, however, the residence time is increased, there is a very rapid increase in the size of the oscillations, but a decrease in their frequency. The stationary-state extent of conversion about which the system is oscillating also changes with τ_{res} . It decreases, corresponding to an increase in α_{ss} , as the residence time becomes longer.

When τ_{res} reaches 312.5, α_3 is no longer unique— α_1 (stable) and α_2 (unstable) become real. However, the oscillations do not immediately die out. There is a very narrow range (Fig. 3b) of τ_{res} over which the limit cycle does not cross the saddle point. If the residence-time is increased to 313, the system crosses the separatrix and the oscillations are abruptly quenched. For any higher residence-times, there is only one realizable solution. This is α_1 , the lowest extent of conversion.

Any subsequent reduction of τ_{res} now leads to a retracing of this lower branch. At $\tau_{res} = 313$, however, the system does not remember that it was oscillatory and the oscillations will not restart. There is a narrow region of hysteresis. The system will not move back onto the limit cycle until α_1 and α_2 coalesce, at $\tau_{res} = 312.5$. When this occurs, the oscillations recommence as large, fully developed pulses. This "hard excitation" leads to a subtle distinction between the mushrooms displayed in Figs. 2 and 3. For the former, the change from stable to unstable focus and, hence, the onset of oscillations occur within the region of stationary-state hysteresis. The "ignition" is, therefore, to a stable stationary-state, not to an oscillatory one.

For this example, the range of residence-time over which hysteresis is exhibited is very small, *ca.* 0.5 of the characteristic timescale. For other values of τ_2 and β_0 , however, it may be more extensive. In the case of a system displaying an isola in the α_{ss} - τ_{res} (or γ_{ss} - τ_{res}) plane, hysteresis is more dramatic. Once the limit cycle surrounding α_3 touches the separatrix of α_2 , the stationary-state extent of conversion falls to the lower branch α_1 . No subsequent variation of τ_{res} whatsoever will cause the oscillatory state to return.

6. PORTRAITS IN THE a - b PHASE PLANE

It is interesting to compare the variety of patterns formed by the trajectories in the a - b phase plane in the present model with those found for the single, irreversible, exothermic reaction proceeding under non-adiabatic conditions. Uppal *et al.* [7] have counted nine different portraits for the latter. All of these can be reproduced, by suitable choices of τ_2 , τ_{res} and β_0 , by our isothermal, autocatalytic system. The nine por-

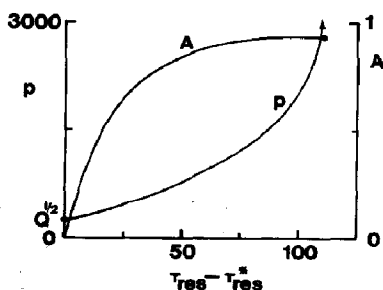


Fig. 7. Dependence of oscillatory period (p) and amplitude (A) on residence-time: $\tau_2 = 40$, $\beta_0 = 1/15$.

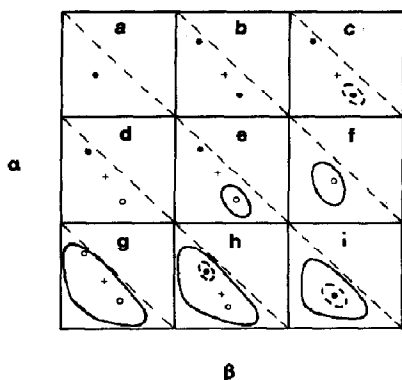


Fig. 8. Variety of phase-portraits displayed by cubic autocatalysis in an isothermal CSTR, when catalyst is not indefinitely stable: ●, stable node or focus; ○, unstable node or focus; +, saddle point; stable limit cycles denoted by full lines, unstable limit cycles denoted by broken lines. Note that in each diagram all singularities lie on the line $\beta = (1 + \beta_0 - \alpha)/(1 + \tau_{res}/\tau_2)$.

traits are shown in Fig. 8. Vaganov *et al.* [8] present a total of 35 possible patterns for the non-isothermal case, some with different combinations of two or more limit cycles. However, they do not give an explicit indication as to whether all of these can be achieved in practice. The present model appears to have almost all of the flexibility of the simple, self-heating system, and so should give rise to an equally extensive selection.

Systems displaying an *isola* in the $\alpha_{ss} - \tau_{res}$ plane do not exhibit all of these portraits. With no catalyst in the inflow, their only unique singularity is a stable node (Fig. 8a), corresponding to ranges of τ_{res} for which the lowest extent of conversion α_1 is unique. For the system represented by Fig. 1, we find a region in which there are two stable stationary-states (α_1 and α_3) and an unstable saddle point (α_2) (Fig. 8b). The uppermost stationary-state α_3 may be surrounded by an unstable limit cycle (Fig. 8c). There is also a region in which α_1 is stable, α_2 a saddle point and α_3 an unstable node or focus (Fig. 8d).

The *mushroom* shown in Fig. 2 displays an additional phase-plot; there is a range of residence times for which the uppermost stationary-state α_3 is unstable and surrounded by a stable limit cycle (Fig. 8e). The *mushroom* in Fig. 3, which corresponds to a higher inlet concentration of catalyst also exhibits a unique, unstable focus surrounded by a stable limit cycle (Fig. 8f).

As the amount of autocatalyst B included in the inflow is increased, the *mushroom* opens out more. It is also then possible for the lowest stationary-state to become unstable. This instability in α_1 is found at the right-hand (long τ_{res}) neck of the *mushroom* as α_2 and α_1 re-emerge and separate. At these residence times the highest stationary-state is also unstable, and a stable limit cycle surrounds all three singularities (Fig. 8g). Further along the lowest branch, α_1 becomes a stable focus, and a second Hopf bifurcation occurs. This gives rise to an unstable limit cycle surrounding α_1 and

lying within the stable limit cycle (Fig. 8h). When the highest and middle stationary-states then coalesce and vanish (at higher τ_{res}), the system displays a unique stable singularity surrounded by two limit cycles (Fig. 8i). As the unstable limit cycle grows in size, it may coalesce with the stable cycle, and sustained oscillatory behaviour then ceases. Ultimately, for $\beta_0 > 1/8$, the two regions of multiplicity and of hysteresis completely disappear. The stationary state diagram has a unique solution for all τ_{res} . Unstable singularities and sustained oscillations persist for these inlet compositions.

7. DEPENDENCE OF STATIONARY-STATES ON RESIDENCE TIME

Figures 1 and 2 (or 3) portray two of the patterns possible for the dependence of the stationary-state extent of conversion on residence-time: representing an *isola* and a *mushroom*, respectively, for each of which there is a loss of "dynamic" stability along the uppermost branch. As the parameters τ_2 and β_0 are varied, so we may encounter different patterns. We have observed a total of eight different forms. Figure 9 shows the bifurcation diagram for our autocatalytic system, presenting the different regions of $\tau_2 - \beta_0$ parameter space in which these patterns arise. A similar diagram has also been given by Escher and Ross [11b].

For high inlet concentration of the catalyst, such that $b_0 > a_0/8$ or $\beta_0 > 1/8$, there is a unique stationary-state at each residence time regardless of the decay time τ_2 . For lower β_0 , the dependence of γ_{ss} on τ_{res} may show either unique or multiple solutions depending on τ_2 . The regions of multiple and unique solutions in Fig. 9 are separated by the lower root of the equation

$$\frac{8}{\tau_2} = (1 + \beta_0)(1 - 2\beta_0) - \frac{1}{4}(1 - 8\beta_0)[3 \pm (1 - 8\beta_0)^{1/2}].$$

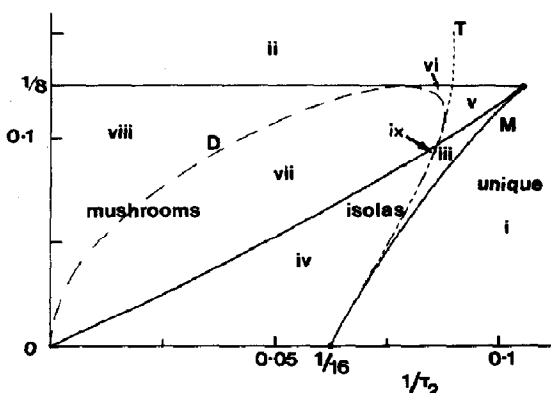


Fig. 9. Bifurcation diagram showing different regions of system parameters β_0 and τ_2 corresponding to different responses for the stationary-state extent of conversion to the residence time: —, curve M separates unique patterns from isolas and mushrooms; —, curve T separates systems exhibiting points of Hopf bifurcation from those which do not; —, curve D corresponds to systems for which the determinant and trace of the Jacobian matrix vanish simultaneously.

(Note that we plot the inverse of the catalyst decay time, τ_2^{-1} , along the abscissa in Fig. 9 for convenience). This leads to the lower section of the curve M , which separates unique patterns (at short τ_2) from isolas (at longer τ_2). The upper root leads to the upper section of the curve M , and separates isolas from mushrooms. The two sections of M meet at a cusp point which has coordinates $\beta_0 = (1/8)$, $\tau_2 = (4^4/3^3)$.

A second curve, labelled T in Fig. 9, represents the marginal conditions for the loss of dynamic stability, i.e. those pairs of β_0 and τ_2 for which the trace of the Jacobian matrix may attain a maximum value of exactly zero as the residence time τ_{res} is varied. For points to the right of this curve (i.e. those with higher values of τ_2^{-1} , corresponding to systems with short-lived catalysts) the trace can never become zero and so Hopf bifurcation to sustained oscillations is not possible. For points to the left of T (i.e. for longer-lived catalysts) the trace of the Jacobian may become positive over some range of residence time, and Hopf bifurcation will occur.

The third curve in Fig. 9, labelled D , corresponds to pairs of the parameters β_0 and τ_2 for which the trace and determinant of the Jacobian matrix vanish simultaneously at the same stationary-state. This lies completely within the region of the multiple solutions, with two solutions τ_2 for any given catalyst inlet concentration, $\beta_0 < 1/8$. With no catalyst in the inflow ($\beta_0 = 0$) we have $Tr = Det = 0$ for $\tau_2 = 16$ and for $\tau_2 \rightarrow \infty$ (i.e. for $\tau_2^{-1} = 0$). With $\beta_0 = 1/8$, the two solutions for τ_2 merge at $\tau_2 = 4^3/3^4 = 12.642$.

In this way, Fig. 9 is divided into 8 clearly distinguishable regions: two of these (i) and (ii) correspond to unique dependences of γ_{ss} on τ_{res} ; two regions, (iii) and (iv), correspond to isolas and; four, (v)–(viii), correspond to mushrooms. Escher and Ross[11b] were the first to discover the patterns (vi) and (viii) and they have also been able to locate a very small region of parameter space containing a third form (ix) of isola. Thus the nine different response patterns for the dependence of the stationary-state extent of conversion on residence time are displayed diagrammatically in Fig. 10, where we use solid lines to represent stable solutions and dashed lines to represent unstable ones.

8. DISCUSSION

The homogeneous autocatalytic reaction scheme studied here was first investigated in general terms by Zel'dovich[9] forty years ago. In its reversible form, it has also been more recently employed by Schlögl[10] as a kinetic model for second-order phase transitions. In that work the concentration of the reactant A is assumed to be held constant in some way, and flow terms are not included. Heinrichs and Schneider[3] adapted Schlögl's analysis to treat the continuous-flow, well-stirred tank reactor. They demonstrated the phenomena of "slowing-down" in these circumstances, in which the relaxation times for the decay of small perturbations could become very long compared with the mean residence-time of the reactor. At the

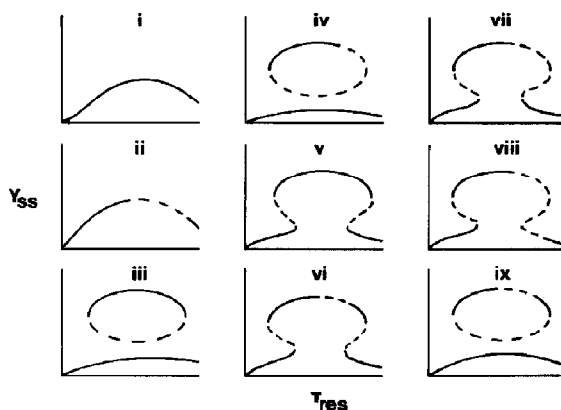


Fig. 10. Different patterns observed for the dependence of stationary-state extent of conversion γ_{ss} on residence-time τ_{res} corresponding to the different regions in Fig. 9: —, stable solutions; ---, unstable solutions.

same time, Lin[2] derived the conditions for multiplicity or for unique solutions. Neither Heinrichs and Schneider nor Lin included the decay reaction $B \rightarrow C$. Consequently their "one variable" systems could not display the complex behaviour and oscillatory solutions discovered by us. Since the publication of our previous paper, we have learned of subsequent, independent studies[11a] by Escher and Ross. They have repeated a part of our earlier work and are in general concordance with our results.

In Figs. 2 and 3 we have marked the amplitude of the oscillations. We have shown both the highest and the lowest extent of conversion achieved during each pulse, and how these extrema vary with the residence time. This seems a more satisfying presentation than that of Uppal *et al.*[7] who showed only the top of the oscillatory amplitude and understate its size. The present approach emphasizes how rapidly the oscillations grow as τ_{res} increases. This is not revealed so clearly if only the upper locus is considered, because the attainment of complete conversion ($\alpha = 0$, $\gamma = 1$) presents a low "ceiling" above which the system cannot go. The lower locus also illustrates how the oscillations may become too large to remain stable. As the amplitude grows, the envelope of oscillations approaches the separatrices of the saddle point. This is reflected in Figs. 1–3 by the minimum extent of conversion achieved during each pulse approaching the branch of solutions γ_2 .

This coalescence of a limit cycle and a saddle point has recently been stressed[12] elsewhere, in a combustion context. It has been proposed as an explanation of the "two-stage" nature of ignition in (non-isothermal) hydrocarbon oxidation. In the present example, this coalescence leads to an extinction of the reaction. The system moves to a state of almost no conversion, the lowest branch in the stationary-state diagram. This means that extinction does not result from "falling-off" the high conversion branch at the right-hand end of the isola or mushroom, as has been indicated elsewhere[1, 9]. Instead it corresponds to a

quite different transition, with the abrupt quenching of an oscillatory state. If the residence-time is subsequently reduced, ignition occurs. The nature of this transition from α_1 to α_3 depends upon the relative positions of τ_{res}^* and the residence time for ignition τ_{ign} at the r.h.s. of the mushroom. If τ_{res}^* is longer than τ_{ign} ($\tau_{res}^* > \tau_{ign}$), the system moves to a steady state, corresponding to a stable (focal) character of α_3 (Fig. 2). If τ_{res}^* is shorter than τ_{ign} , there is ignition from a steady state to an oscillatory state, corresponding to a stable limit cycle surrounding α_3 (Fig. 3). The ignition and extinction at the left-hand end of the mushroom or isola, are of the more familiar type between two stable and steady stationary-states.

We have already touched on some of the similarities and differences between our present model and the non-isothermal, exothermic reaction



Dimensionless forms for the mass and heat balance equations are

$$\begin{aligned} \frac{d\alpha}{d\tau} &= -\alpha \exp\left(\frac{\theta}{1+\epsilon\theta}\right) + \frac{(1-\alpha)}{\tau_{res}} \\ \frac{d\theta}{d\tau} &= \theta_{ad} \alpha \exp\left(\frac{\theta}{1+\epsilon\theta}\right) - \left(\frac{1}{\tau_N} + \frac{1}{\tau_{res}}\right)\theta \end{aligned} \quad (8)$$

where $\epsilon = RT_a/E$ is commonly assumed to be very small, $\epsilon \rightarrow 0$. The group of terms represented by θ_{ad} is the maximum dimensionless temperature-excess that is achieved when all the reactant A is consumed under adiabatic conditions

$$\theta_{ad} = \frac{(-\Delta H)a_0 E}{c_p \sigma RT_a^2}$$

and corresponds to the parameter B used by Uppal *et al.* [7].

For the autocatalytic reaction scheme, the governing equations are

$$\begin{aligned} \frac{d\alpha}{d\tau} &= -\alpha\beta^2 + \frac{(1-\alpha)}{\tau_{res}} \\ \frac{d\beta}{d\tau} &= \alpha\beta^2 - \left(\frac{1}{\tau_2} + \frac{1}{\tau_{res}}\right)\beta. \end{aligned} \quad (9)$$

Both pairs contain a non-linear source term; an exponential temperature-dependence on θ for the non-isothermal case, and a quadratic dependence on β for our model. The rates R of chemical production of heat or autocatalyst B as functions of the extent of conversion $\gamma = 1 - \alpha$ in a stationary-state may be expressed, respectively, by

$$R_{sh} = \theta_{ad}(1-\gamma) \exp\left\{\frac{\theta_{ad}\gamma_{ss}}{[1+(\tau_{res}/\tau_N)]}\right\} \quad \text{self-heating}$$

and

$$R_{ia} = \frac{(1-\gamma_{ss})\gamma_{ss}^2}{[1+(\tau_{res}/\tau_2)]^2} \quad \text{isothermal autocatalysis.}$$

These "generation curves", shown in Fig. 11, have a similar general shape. Both rise slowly at first from their values at $\gamma = 0$, have an inflexion point and then reach a maximum before falling to zero as conversion becomes complete ($\gamma \rightarrow 1$).

Generation curves of similar overall shape are also found for some enzyme systems and are exhibited by surface reactions obeying Langmuir-Hinshelwood kinetics. In the solution phase the isothermal cubic rate-law has been shown to approximate the behaviour observed in the reactions between arsenite and iodate [13] and between permanganate and oxalate ions [14]. For both non-isothermal and autocatalytic systems the "loss line" L is represented by a straight line whose gradient is determined by τ_{res} and by τ_2 . The stationary-states correspond to the intersections of R and L . Multiplicity arises or disappears when R and L become tangential. If the autocatalytic reaction is operated with a non-zero inlet concentration of B , both R curves may have two tangencies; the upper tangency corresponds to a relatively steep loss-line, the lower tangency to a rather flat one. Each tangency condition is satisfied by two residence-times, thus we may have four such bifurcation points. These circumstances lead to mushrooms in the $\gamma - \tau_{res}$ plane. The loss-line has a minimum slope when $\tau_{res} = \tau_2$, when the gradient is $4/\tau_2$ in the scale of Fig. 11. If this line of minimum slope lies above the lower tangent, an isola is found.

The Arrhenius rate-law allows the temperature excess, arising from heat released in the exothermic step, to play an autocatalytic role similar to that of θ_{ad} . Newtonian cooling at the vessel walls, which has a rate proportional to θ , has an analogous effect to our first-order decay of catalyst. Adiabatic operation, with no heat-loss at the walls, corresponds to the timescale for cooling τ_N becoming infinite and, is equivalent to infinite catalyst stability ($\tau_2 \rightarrow \infty$). If there is no heat-loss, isolas and mushrooms are not

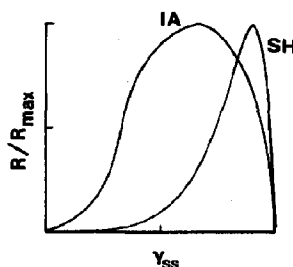


Fig. 11. Dependence of rate of heat generation or autocatalyst production on the stationary-state extent of conversion γ_{ss} for first-order, irreversible exothermic reaction SH and for cubic autocatalysis IA respectively.

found, nor are oscillations, because the system has only one independent variable.

An additional application of our rate-law, therefore is as a non-isothermal system with the Arrhenius temperature-dependence replaced by a quadratic one. Such an approximation has been used to great advantage[15] in thermal explosion theory.

The isothermal, autocatalytic system does not recreate the full range of complexity of the non-isothermal first-order reaction. In particular, the simple form considered here does not display all of the patterns of stationary-state dependence on residence time found[7, 16] for the latter. One important source of the additional richness in the non-isothermal system is the parameter θ_{ad} . The numerical value of this dimensionless group may be varied almost indefinitely, from very high values $\cong 10^3$ to low values of ca. 3 or 4, by diluting the system (thus increasing the specific heat capacity). For our autocatalytic system, the analogous factor arising in (9) relates to the stoichiometry of the step $A + 2B \rightarrow 3B$. This has the value unity. We may imagine a more general form



The differential equation for β would then have the form

$$\frac{d\beta}{d\tau} = (m - n)\alpha\beta^2 - \left(\frac{1}{\tau_2} + \frac{1}{\tau_{res}}\right)\beta.$$

There are, however, only a very limited number of possibilities for m and n that are chemically reasonable.

CONCLUSIONS

(1) The simple, cubic autocatalytic reaction scheme studied here gives rise to nine different patterns for the dependence of stationary-state extents of reactant conversion on residence time. These include closed curves or isolas, and mushrooms.

(2) For the simplest circumstance, in which there is no catalyst included in the inflow, the lowest branch of stationary-states, corresponding to small extents of conversion is always stable: the middle branch is always unstable. The character and stability of the highest branch varies with the residence-time and with the catalyst stability.

(3) The loss of stability along the upper branch is accompanied by a Hopf bifurcation. This may be either subcritical, leading to unstable limit cycles surrounding stable stationary-states, or supercritical, leading to stable limit cycles and sustained oscillatory responses.

(4) When some autocatalyst is included in the

inflow, yet richer behaviour is found. The lowest branch of stationary-states may change character, to focus from node. It may also lose its stability, leading to a second Hopf bifurcation.

(5) This richer behaviour leads to more complicated phase-plane portraits: a stable limit cycle may surround three unstable stationary-states, or stable and unstable limit cycles may now coexist.

(6) The nature of the Hopf bifurcation and stability of the limit cycles have been established directly by analytical evaluation of the Floquet exponents and by extensive numerical experiment.

(7) The amplitude and period of sustained oscillatory both increase as the residence time of the system is lengthened. If the amplitude of the oscillations becomes too large, the limit cycle merges with the separatrices of the unstable saddle point. This leads to an abrupt quenching or extinction of the oscillatory state. This differs from the conventional mode of extinction, which corresponds to "falling-off" the end of the isola or mushroom.

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